Friction on Different Surfaces

Friction between an object and the surface it is on changes depending on the surface and its roughness.

Aim: To test the friction on different surfaces

Equipment: Wooden block
1 x 1 Newton spring balance
1 x 5 Newton spring balance
1 x 10 Newton spring balance
Various surfaces to test

Diagram of Equipment Set-up

Hypothesis:

Method:
1. Calibrate each spring balance by holding it up and lining up the zero line with the top of the spring.
2. Place the wooden block flat on the first surface.
3. Hook the spring balance through the ring on the wooden block.
4. Gently pull the wooden block across the surface at a steady speed.
5. Record your results in the table below.
6. Repeat twice more on the same surface.
7. For each new surface, repeat steps 1 to 6.

Results:

Surface	Trial 1	Trial 2	Trial 3	Average
Sandpaper				
Tiles				
Lino				
Rubber				
Carpet				
Wood				
Desk				
Grass				
Sand				

Questions:
1. Which surface had the most friction?
2. Which surface had the least friction?
3. Do your results match your hypothesis?
4. How could you make your experiment more accurate?